

Chapter 9

Heredity and Evolution

1. The transmission of characters from parents to offspring is known as:

- A) Variation
- B) Evolution
- C) Heredity
- D) Adaptation

Answer: C) Heredity

2. The study of heredity and variation is called:

- A) Cytology
- B) Genetics
- C) Morphology
- D) Physiology

Answer: B) Genetics

3. Who is known as the Father of Genetics?

- A) Charles Darwin
- B) Gregor Mendel
- C) Hugo de Vries
- D) Lamarck

Answer: B) Gregor Mendel

4. Mendel conducted his experiments on:

- A) Maize plant
- B) Pea plant
- C) Wheat plant
- D) Mustard plant

Answer: B) Pea plant

5. Mendel studied inheritance of traits in pea plants using:

- A) Flowers
- B) Seeds
- C) Stem length
- D) All of these

Answer: D) All of these

6. A gene is a unit of:

- A) Heredity
- B) Chromosome
- C) DNA
- D) Protein

Answer: A) Heredity

7. Genes are made up of:

- A) DNA
- B) RNA
- C) Protein
- D) Chromatin

Answer: A) DNA

8. The alternative forms of a gene are called:

- A) Factors
- B) Alleles
- C) Characters
- D) Chromosomes

Answer: B) Alleles

9. In pea plant, tallness (T) is dominant over dwarfness (t). The genotype of a tall heterozygous plant is:

- A) TT
- B) tt
- C) Tt
- D) tTt

Answer: C) Tt

10. The physical expression of a gene is called:

- A) Genotype
- B) Phenotype
- C) Character
- D) Allele

Answer: B) Phenotype

11. The genetic makeup of an organism is its:

- A) Phenotype
- B) Genotype
- C) Allele

D) Chromosome

Answer: B) Genotype

12. In monohybrid cross, the F_2 generation shows a phenotypic ratio of:

A) 1:2:1

B) 3:1

C) 9:3:3:1

D) 1:1

Answer: B) 3:1

13. The genotypic ratio in F_2 generation of a monohybrid cross is:

A) 1:2:1

B) 3:1

C) 9:3:3:1

D) 1:1

Answer: A) 1:2:1

14. In dihybrid cross, the phenotypic ratio obtained by Mendel was:

A) 1:2:1

B) 3:1

C) 9:3:3:1

D) 1:1

Answer: C) 9:3:3:1

15. Sex of a child is determined by:

A) Mother's chromosome

B) Father's chromosome

C) Both equally

D) Hormones

Answer: B) Father's chromosome

16. Human males have which sex chromosomes?

A) XX

B) XY

C) YY

D) XO

Answer: B) XY

17. Human females have which sex chromosomes?

- A) XY**
- B) XX**
- C) YY**
- D) XO**

Answer: B) XX

18. The number of pairs of chromosomes in human cells is:

- A) 23**
- B) 24**
- C) 22**
- D) 46**

Answer: A) 23

19. Variations are important for:

- A) Growth**
- B) Evolution**
- C) Respiration**
- D) Photosynthesis**

Answer: B) Evolution

20. The process by which new species arise from existing ones is called:

- A) Heredity**
- B) Reproduction**
- C) Evolution**
- D) Mutation**

Answer: C) Evolution

21. Who proposed the theory of evolution by natural selection?

- A) Mendel**
- B) Lamarck**
- C) Charles Darwin**
- D) Hugo de Vries**

Answer: C) Charles Darwin

22. Which of the following shows the occurrence of evolution?

- A) Formation of new species**
- B) Inheritance of acquired characters**
- C) Use and disuse of organs**

D) Both A and B

Answer: A) Formation of new species

23. The evidence of evolution from fossils is known as:

- A) Comparative anatomy
- B) Paleontological evidence
- C) Embryological evidence
- D) Genetic evidence

Answer: B) Paleontological evidence

24. The human hand, whale's flipper, and bat's wing are examples of:

- A) Analogous organs
- B) Homologous organs
- C) Vestigial organs
- D) Mutated organs

Answer: B) Homologous organs

25. The wings of a bat and a bird are examples of:

- A) Homologous organs
- B) Analogous organs
- C) Vestigial organs
- D) Mutated organs

Answer: B) Analogous organs

26. The appendix in humans is an example of:

- A) Homologous organ
- B) Analogous organ
- C) Vestigial organ
- D) Functional organ

Answer: C) Vestigial organ

27. The appearance of new characters in offspring not present in parents is called:

- A) Variation
- B) Mutation
- C) Heredity
- D) Evolution

Answer: A) Variation

28. Evolution of life is supported by:

- A) Fossils**
- B) Embryology**
- C) Morphology**
- D) All of these**

Answer: D) All of these

29. Which of the following is an example of acquired character?

- A) Eye colour**
- B) Hair colour**
- C) Muscular body of a wrestler**
- D) Blood group**

Answer: C) Muscular body of a wrestler

30. The first form of life appeared about:

- A) 500 years ago**
- B) 5000 years ago**
- C) 3.5 billion years ago**
- D) 3.5 million years ago**

Answer: C) 3.5 billion years ago
